

ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems.

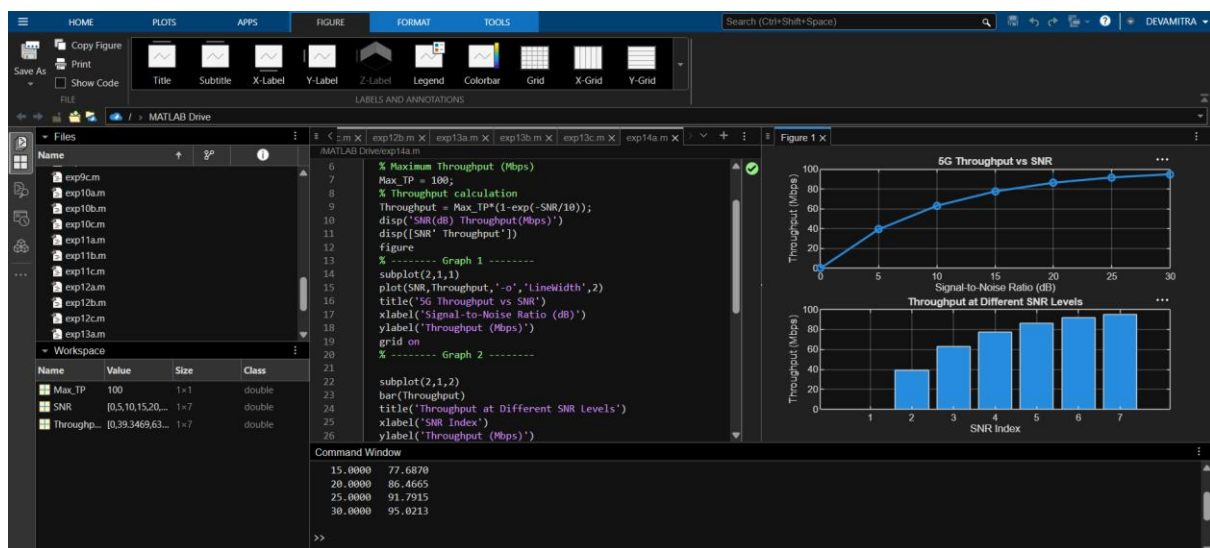
Ex.No. 14: 5G Throughput vs SNR with Modulation Labels Date

Objective 1 Matlab Code and Output:

Program:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation
Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,Throughput,'-o','LineWidth',2)
title('5G Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
bar(Throughput)
title('Throughput at Different SNR Levels')
xlabel('SNR Index')
ylabel('Throughput (Mbps)')
grid on
```

Output:



Objective 2 Matlab Code and Output:

Program:

```
clc;

clear;

close all;

% SNR values (dB)
SNR = 0:5:30;

% Simulated Bit Error Rate (BER)
BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];

% Reliability (%)
Reliability = (1-BER)*100;

disp('SNR(dB) BER Reliability(%)')
disp([SNR' BER' Reliability'])

figure

% ----- Graph 1 -----
subplot(2,1,1)
semilogy(SNR,BER,'-o','LineWidth',2
) title('SNR vs Bit Error Rate in 5G')
xlabel('Signal-to-Noise Ratio (dB)')
```

```

ylabel('Bit Error Rate (BER)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

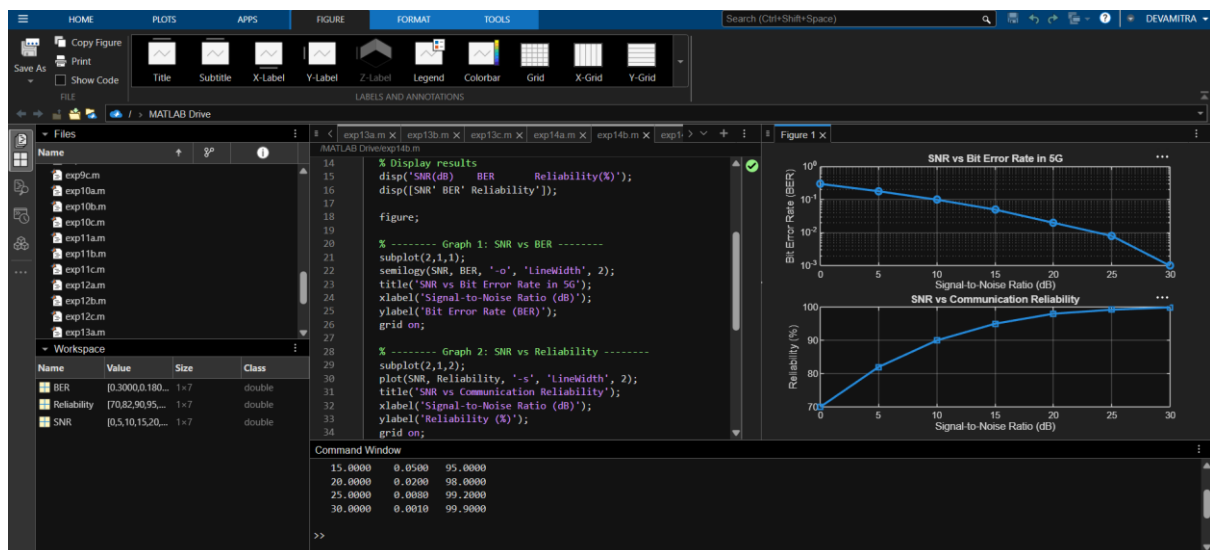
plot(SNR,Reliability,'-s','LineWidth',2)

title('SNR vs Communication
Reliability') xlabel('Signal-to-Noise Ratio
(dB)') ylabel('Reliability (%)')

grid on

```

Output:



Objective 3 Matlab Code and Output:

Program:

```

clc;

clear;

close all;

% Modulation schemes
Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};

% Required SNR (dB)

```

```
SNR = [5 10 18 25];

% Spectral Efficiency (bits/s/Hz)

Efficiency = [2 4 6 8];

disp('Required SNR (dB) and Spectral Efficiency')

disp(table(Modulation',SNR',Efficiency', ...

    'VariableNames',{'Modulation','SNR_dB','Efficiency'}))
```

figure

```
% ----- Graph 1 -----
```

```
subplot(2,1,2)
```

```
plot(SNR,Efficiency,'-o','LineWidth',2)
```

```
title('SNR vs Spectral Efficiency')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Spectral Efficiency (bits/s/Hz)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,1)
```

```
plot(1:4,SNR,'-s','LineWidth',2)
```

```
xticks(1:4)
```

```
xticklabels(Modulation)
```

```
title('Required SNR for Different Modulation Schemes')
```

```
xlabel('Modulation Scheme')
```

```
ylabel('Required SNR (dB)')
```

```
grid on
```

Output:

